Leukemia Classification Using Gene Expression Data

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Introduction

Leukemia, a group of blood cancers arising from immature blood cells in the bone marrow, is a severe and life-threatening disease. Because different leukemia subtypes exhibit distinct clinical symptoms and therapeutic responses, accurate subtype classification is essential for selecting appropriate treatments and improving patient outcomes. Advances in molecular biology have revealed strong links between gene dysregulation and cancer, highlighting the importance of understanding how abnormal gene expression – whether overactivation or insufficient activation – contributes to disease behavior.

Therefore, Gene expression profiles provide valuable information for clinical diagnosis, particularly in distinguishing leukemia subtypes. In this project, we analyze a gene expression dataset with the aim of classifying two major leukemia types: Acute Lymphoblastic Leukemia (ALL) and Acute Myeloid Leukemia (AML). We further seek to identify the most informative genes that could serve as potential biomarkers or targets for future research. At the same time, this project provides an opportunity to apply both parametric and non-parametric classification methods and to follow a typical workflow for analyzing high-dimensional biological data.

Data Analysis

The dataset consists of gene expression measurements from 72 patients, including 47 diagnosed with Acute Lymphoblastic Leukemia (ALL) and 25 with Acute Myeloid Leukemia (AML). The response variable is binary, indicating the leukemia subtype, while the predictors are the expression levels of 3,571 genes, which are continuous and centered around zero. To explore the predictive power of these high-dimensional features, we apply both parametric and non-parametric classification methods, as well as dimensionality reduction via principal component analysis (PCA), individually and in combination, with the intention to evaluate classification performance and identify the most informative genes.

1. Parametric Methods

Common parametric methods, including logistic regression, linear discriminant analysis (LDA), and Naive Bayes, rely on assumptions about the structure of the data, such as normality, uncorrelated predictors, and equal covariance matrices across groups. To assess these assumptions, we first plotted histograms of all predictors within each leukemia subtype. Both ALL (Fig.1.1) and AML (Fig.1.2) subgroups exhibited approximately symmetric, bell-shaped distributions centered around zero, suggesting that the normality assumption is roughly satisfied.

Next, we evaluated the correlation among predictors, which is an assumption for logistic regression and Naive Bayes. Given the high dimensionality of the dataset, we focused on the 10 predictors with the largest variances, as they contribute most to the variation in the response. Several pairs of predictors displayed strong collinearity; for example, G2213 and G3091 had a correlation of 0.987, clearly violating the uncorrelatedness assumption. We also examined the covariance matrices for LDA. The covariance structure for the ALL group generally exhibited higher magnitudes, with several values exceeding 1, whereas the AML matrix entries were smaller and often of opposite sign. For instance, G3019 and G1009 had a

covariance of 1.043 in the ALL sample but only -0.049 in AML. This indicates that the assumption of similar covariance matrices across groups is not met.

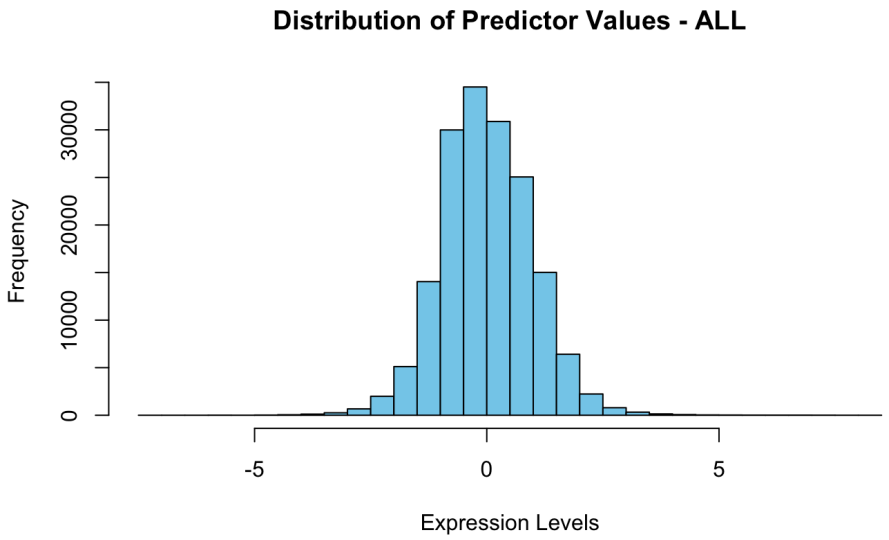

Fig.1.1 Histogram of Genes within ALL Samples

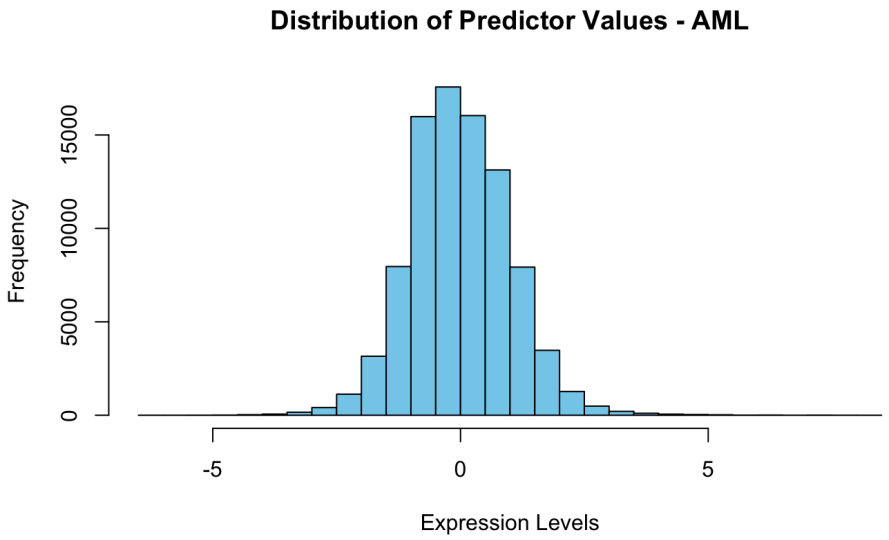

Fig.1.2 Histogram of Genes within AML Samples

Taken together, these diagnostic checks indicate that the parametric assumptions underlying logistic regression, LDA, and Naive Bayes are substantially violated in this dataset. Consequently, we proceeded to explore non-parametric classification methods.

2. Non-parametric Methods

Non-parametric methods, including classification trees, bagging of trees, random forest, and AdaBoost, have become the optimal methods that we can start with. The classification tree utilizes G2481 as the root, with no internal nodes and only two leaves. This classification tree is highly interpretable with a low misclassification rate of 0.13, but a single gene threshold ignores most predictors and multigene interactions. The K-Nearest Neighbor

(KNN) method outputs a smaller misclassification rate of 0.067 and indicates that k is optimal at 4. However, KNN is restricted in terms of interpretability, which contributes a little to our conclusion of biological means.

We also assemble classification trees into bags and random forests to avoid overfitting by boostrapped samples. In the bagging of trees, we generate 57 different bootstrapped training sets and fit a decision tree on each training set. The overall prediction is the most commonly occurring majority group among all 57 predictions. Random forests split the bootstrapped sample, each time depending on a random sample of 59 ($\sqrt{3571}$) and using only one of these 59 predictors. Compared to the bagging of trees, each decision in the random forest considers a subset of predictors, which averages out the effects of the most important predictors on the decision rule. Although the random forest is an improvement based on the bagging, the misclassification rates of these two methods are the same at 0.067. This can be attributed to the small size of the test set and the possibility that a few dominant predictors, such as G2481, reduce the benefit of additional feature randomization. We will investigate the reasons behind this in the later sections. In addition to the bagging and random forests, adaptive boosting (AdaBoost) is another method that combines the predictions of multiple weak models into a weighted ensemble, placing more emphasis on observations that previous models misclassified. More specifically, trees are grown sequentially in the rule that observations that have been misclassified the most are given the highest weight in subsequent trees. Each tree contributes to the classification rule proportionally to its overall prediction accuracy. As a result, AdaBoost also misclassifies 6.7% of the samples from the test set. In terms of prediction accuracy, we are unable to conclude which non-parametric method is the best; however, these methods offer valuable insights into how the most important predictors significantly contribute to the decision rules.

In the bagging and random forests, MeanDecreaseGini measures how much a predictor variable helps improve the purity of classification at each split. A high Gini decrease demonstrates that splitting on that gene consistently produces nodes with more clearly separated classes, indicating that the gene is important for distinguishing between the two leukemia types. When we constructed the MeanDecreaseGini plots and extracted the exact values, we found that after the 5th most important predictor, the decrease in Gini dropped significantly. This implies that only a small subset of predictors influences the classification the most, whereas the others play trivial roles. Based on this observation, the 10 most important predictors are selected from each method. In AdaBoost, importance scores are assigned based on how much each gene contributes to reducing errors across the boosting iterations. In this case, we select the 10 predictors with the highest scores. The 10 predictors obtained from three methods through distinctive criteria overlap significantly with each other.

Compared to the bagging and AdaBoost, the predictors obtained from the random forest are more reliable. While the bagging simply builds multiple decision trees using different bootstrapped samples of the training data and uses all predictors at every split, the random forest introduces randomness by selecting a random subset of predictors at each split. This prevents dominant variables from appearing in every tree, allowing the model to better assess the relative importance of all features and reducing bias toward predictors with high variance. Moreover, instead of focusing on correcting errors by reweighting misclassified observations and can overfit noisy observations in AdaBoost, the random forest emphasizes generalizable patterns, leading to more robust estimates of predictor importance. In this case, we choose to use the 10 most important predictors selected from the random forest (Fig. 2):

$$G2481\ G3441\ G1652\ G979\ G435\ G436\ G2226\ G2789\ G1249\ G3216$$

These predictors are crucial to be brought into Principal Component Analysis (PCA), and uncorrelated principal components satisfy the assumptions of parametric methods, that we expand our model choices.

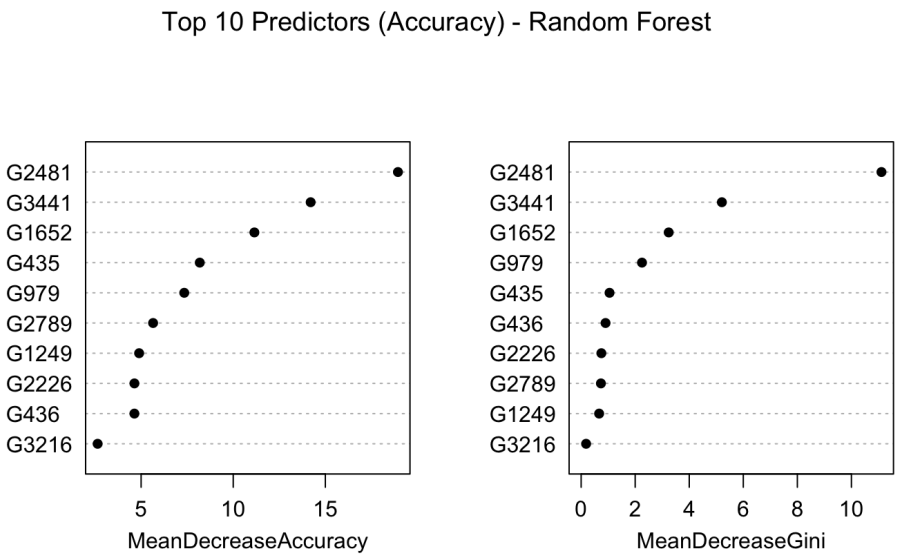

Fig.2 10 Most Important Predictors in Random Forest

3. PCA and Combination of Methods

To further reduce dimensionality and address multicollinearity among the predictors, we applied principal component analysis (PCA) to the top 10 genes identified by the random forest-based predictor selection step. A preliminary boxplot assessment confirmed that these genes have comparable variance, so additional scaling was unnecessary. The scree plot (Fig.3) showed a clear elbow at the second component, and the cumulative proportion of variance explained reached 0.7885 by PC2, indicating that the first two components capture the majority of the structure in the data. We therefore retained PC1 and PC2 for downstream analysis.

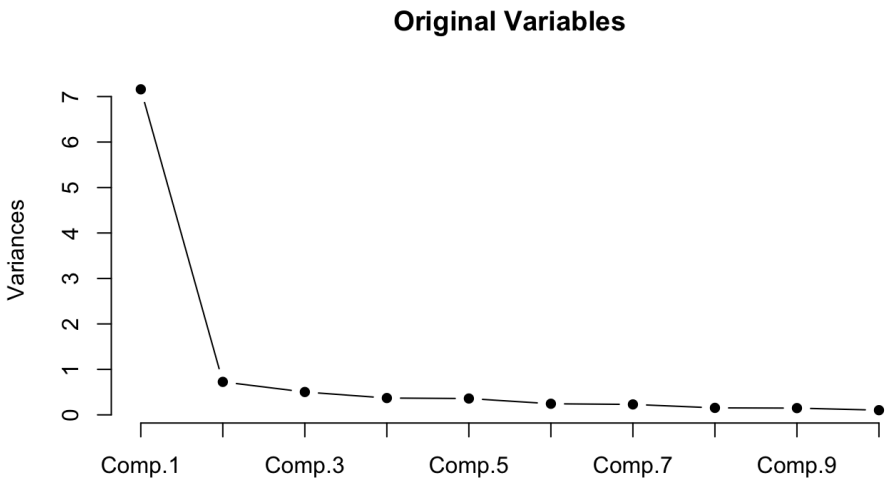

Fig.3 Scree Plot of The Principal Component Analysis (PCA)

To interpret the components, we visualized their loadings in a barplot (Fig.4) and a scatterplot (Fig.5). For PC1, three genes (G3441, G435, and G2789) exhibited positive loadings, while the remaining seven genes had negative loadings, all with magnitudes around 0.3. For PC2, six genes (G3441, G2226, G435, G436, G2789, and G1249) showed positive loadings ranging from 0.2 to 0.6, whereas the other four genes loaded close to zero. The two-dimensional loading scatterplot provided a clearer view of how each gene contributes jointly to PC1 and PC2: G3441, G435, and G2789 loaded positively on both components; G1249, G436, and G2226 loaded negatively on PC1 but positively on PC2; and the remaining four genes (G2481, G1652, G3216, G979) loaded negatively on PC1 with near-zero contribution to PC2. These patterns suggest that the derived components are both interpretable and effective at summarizing gene-level variation. Because our analysis does not rely on prior biological knowledge of the genes, and because the loadings already display adequate sparsity and separation, we did not pursue sparse PCA.

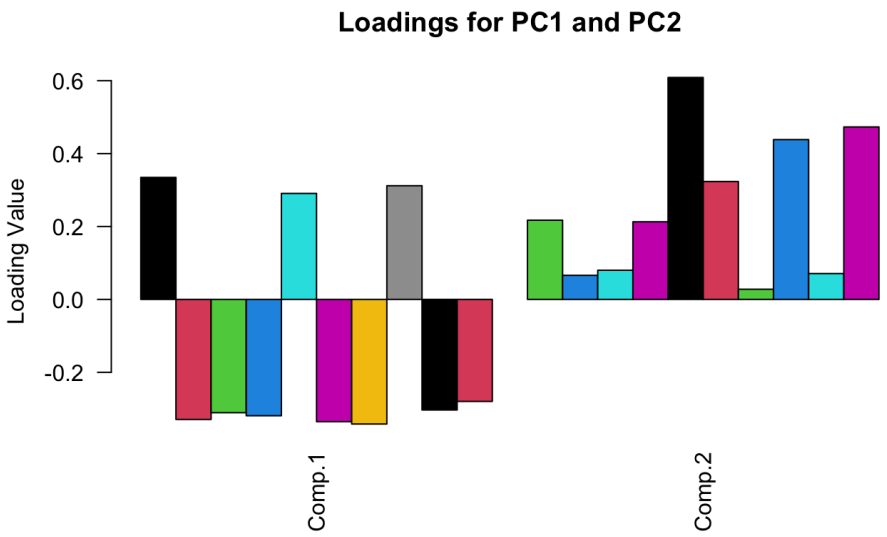

Fig.4 Bar Plot of the Loadings for The First Two Principal Components (PC1 and PC2)

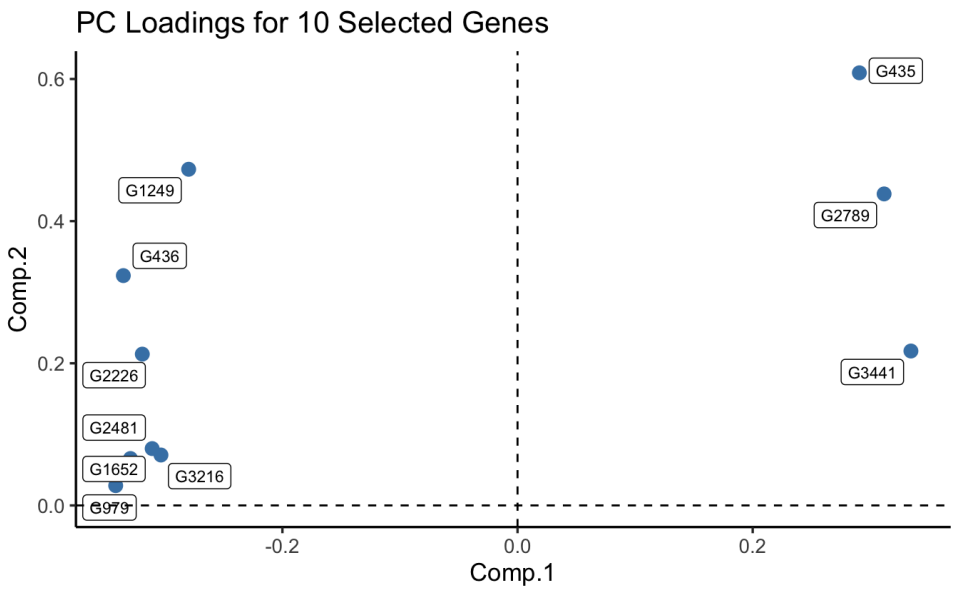

Fig.5 Scatter Plot of the Loadings for The First Two Principal Components (PC1 and PC2)

With the issue of predictor collinearity addressed, we next applied the first two principal components as inputs to parametric classification models. By using the PCs as predictors, we ensured that the assumptions of uncorrelated predictors (for logistic regression and Naive Bayes) and approximately equal covariance matrices (for LDA) are more closely satisfied, while retaining the majority of the variance in the data.

In Naive Bayes, we built a model on the training set to estimate the class probabilities and the conditional distributions of PC1 and PC2 within each leukemia subtype. Using this model, we then predicted the class labels for the test set based on the observed PC1 and PC2 scores. The Naive Bayes classifier correctly classified 14 out of 15 test samples, resulting in a misclassification rate of approximately 6.7%, indicating that the PCs effectively captured the information needed to distinguish between ALL and AML. In LDA, we fit a linear discriminant function to the same training set, using PC1 and PC2 to compute group means and the linear combination of predictors that maximally separates the two subtypes. Applying this discriminant function to the test set, LDA correctly classified all 15 samples, achieving a misclassification rate of 0%. This demonstrates that the linear separation in the reduced PC space is sufficient for perfect discrimination in this dataset.

In logistic regression, we modeled the probability of each leukemia subtype as a logistic function of PC1 and PC2. The model estimates the log-odds of being classified as AML versus ALL as a linear combination of the two principal components. In our fitted model, the estimated coefficient for PC1 was -4.806 and for PC2 was 1.842, while the intercept was -0.832. These estimates indicate that higher values of PC1 are associated with lower odds of AML, whereas higher values of PC2 slightly increase the odds, though individually the coefficients were not statistically significant (p-values 0.179 and 0.498, respectively), likely due to the small sample size. Using the training set, predicted log-odds for the test samples were converted into probabilities and then into class labels using a 0.5 threshold. On the test set, the logistic regression classifier correctly classified 14 out of 15 samples, producing a misclassification rate of 6.7%. This result, consistent with Naive Bayes, demonstrates that PC1 and PC2 effectively capture the relevant variation for classification and provide interpretable contributions to the odds of each leukemia subtype.

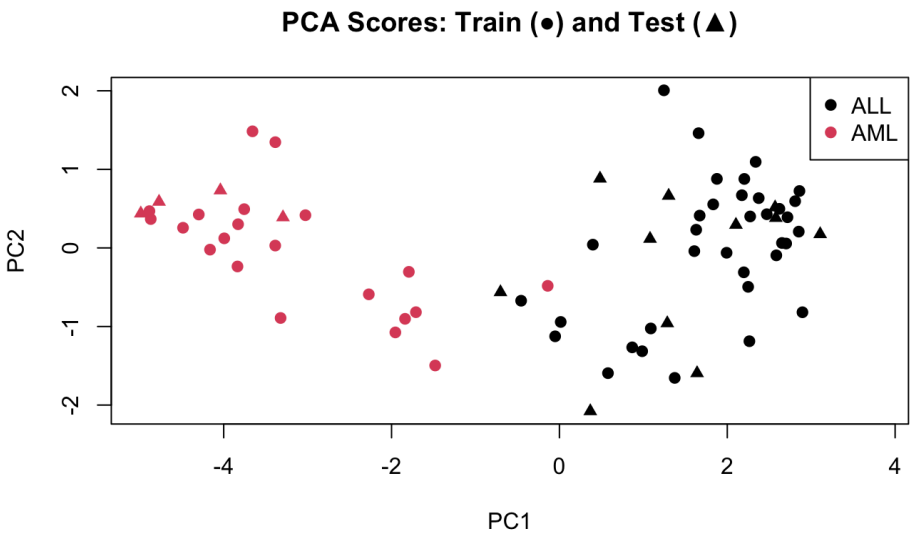

Fig.6 PCA Score Plot: The Distribution of AML (Red) and ALL (Black) Samples

To take an intuitive look at how the samples are distributed along PC1 and PC2, we plotted the PCA scores for all training and test samples. In this plot (Fig.6), training points are represented as circles and test points as triangles, while colors indicate leukemia subtype, with ALL in black and AML in red. The visualization reveals a clear separation between the ALL and AML clusters, primarily along the PC1 axis: samples with positive PC1 scores generally correspond to ALL, whereas those with negative PC1 scores correspond to AML. A few points overlap near the center, reflecting cases that are less clearly separated. Notably, one black triangle (ALL test sample, PC1 = -0.70, PC2 = -0.56) lies near the AML cluster, corresponding to the single misclassification observed in both the Naive Bayes and logistic regression models.

Discussion

From a biological perspective, our results suggest that the distinction between ALL and AML is driven by coordinated patterns of gene expression rather than isolated effects of individual genes. The first principal component captures a dominant transcriptional axis that almost perfectly separates the two subtypes, consistent with differences in underlying cellular lineage and regulatory programs. Genes with large loadings on PC1 may serve as potential biomarkers for subtype classification or as candidates for further functional investigation. In particular, genes such as G3441, G435, and G2789, which have strong positive loadings on PC1, are likely to be positively associated with ALL and negatively associated with AML. Conversely, genes including G2481, G1652, G3216, and G979, which load negatively on PC1, appear to be positively associated with AML and negatively associated with ALL.

Since the misclassification rates generated from each model appear to be similar to each other, we cannot make any final conclusion on which model is the best for this data set, but this suggests that if we collect more data to further evaluate the models, we can select the best ones out of them. The presence of a small number of overlapping or misclassified samples highlights the inherent biological heterogeneity of leukemia and suggests that some patients may exhibit intermediate or atypical expression profiles. Overall, this analysis demonstrates that combining gene selection, dimensionality reduction, and statistical classification provides an effective framework for extracting biologically meaningful insights from high-dimensional genomic data.